A PROJECT REPORT

ON

P_311: KNOWLEDGE-DRIVEN IoT FAULT DIAGNOSIS ASSISTANT FOR INDUSTRIAL ROTATING MACHINERY

Submitted in partial fulfillment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE AND ENGINEERING / INTERNET OF THINGS

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[NAME OF INSTITUTION / COLLEGE / UNIVERSITY]

[City, State, PIN Code]

Academic Year: 2025 – 2026

CERTIFICATE

This is to certify that the project report entitled **"P_311: Knowledge-Driven IoT Fault Diagnosis Assistant for Industrial Rotating Machinery"** is a bona fide record of the work carried out by:

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in partial fulfillment for the award of the degree of **Bachelor of Technology in Computer Science and Engineering / Internet of Things** at **[Name of Institution / University]** during the academic year **2025–2026**.

The work reported herein has been audited and verified to meet academic engineering standards.

_____

External Examiner Signature: _____

Date of Examination: `[DD / MM / YYYY]`

DECLARATION

We hereby declare that the project work entitled **"P_311: Knowledge-Driven IoT Fault Diagnosis Assistant for Industrial Rotating Machinery"** submitted to the **Department of [Department Name], [Institution Name]**, is a record of original engineering work done by us under the supervision of **[Guide Name]**, Department of `[Department Name]`.

We further confirm that:

1. The technical results, software code, and evaluations presented herein are our own and accurately reflect the system implementation.
2. In accordance with university ethics, physical electric motors were not physically damaged in the laboratory; realistic physical dynamics were executed via a mathematical digital twin simulator.
3. This report has not been submitted previously to any other university or institution for the award of any degree or diploma.

Place: `[City Name]`

Date: `[DD / MM / YYYY]`

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ABSTRACT

In modern industrial facilities, 3-phase induction motors drive the vast majority of critical rotating machinery, including centrifugal pumps, air compressors, conveyor lines, and industrial fans. Unexpected mechanical and thermal failures—predominantly bearing raceway spalling (41–51%) and stator insulation burnout (28–36%)—cause catastrophic production line stoppages, with unplanned downtime costs ranging from $20,000 to $250,000 per hour. While traditional Supervisory Control and Data Acquisition (SCADA) systems monitor simple binary thresholds, they offer zero diagnostic context, induce operator alarm fatigue, and require technicians to manually search equipment binders. Furthermore, modern cloud-based AI predictive platforms are hindered by data privacy risks, network bandwidth constraints, recurring subscription costs, and the inability to operate in air-gapped industrial environments.

This project presents **Project P_311: Knowledge-Driven IoT Fault Diagnosis Assistant**, an edge-native, explainable AI IoT monitoring architecture engineered for 3-phase induction motors. The system decouples high-throughput 1 Hz telemetry ingestion over MQTT (Eclipse Mosquitto port 1883) from a sub-millisecond in-memory deterministic fault detection engine. The safety detector adheres strictly to **ISO 10816-3** vibration severity standards (Zone C warning at 4.5 mm/s, Zone D trip at 7.1 mm/s), **IEC 60034-1** Class F stator thermal insulation limits (trips at 95°C), and full-load ampere slip constraints. To eliminate black-box opacity and factual hallucination, the system couples a local CPU-bound Retrieval-Augmented Generation (RAG) pipeline (`sentence-transformers/all-MiniLM-L6-v2`, 384 dimensions) with a quantized local Large Language Model (`qwen2.5:1.5b-instruct` in Ollama). The RAG pipeline queries 34 pre-indexed technical manual chunks in $< 1\text{ ms}$, grounding the LLM's synthesis of actionable maintenance checklists, probable root causes, safety Lockout/Tagout (LOTO) protocols, and verified document citations.

To ensure non-stop industrial reliability, the architecture features a dual-database persistence engine (PostgreSQL 16 in Docker with native JSONB, automatically falling back to WAL-mode SQLite if Docker is offline) and a deterministic rule-based fallback generator that guarantees diagnostic synthesis even if the local LLM times out or is offline. System visualization is segregated into an operational Grafana SCADA time-series monitor (port 3000) and an interactive React 18 + Vite engineering dashboard (port 8000) featuring Server-Sent Events (SSE) streaming, an animated SVG Digital Twin motor, audio alarm annunciator, printable maintenance work orders, and an interactive Viva AI tutor chat.

Empirical evaluation over a curated 4,000-sample industrial dataset demonstrated an overall fault detection accuracy of **99.38%** (Macro F1: 0.9931), a RAG Mean Reciprocal Rank (MRR) of **1.000** (Hit@1: 100%), sub-millisecond rule evaluation (0.84 ms), and a local memory footprint under **1.35 GB VRAM** and **1.2 GB RAM**, proving that fully explainable, cloud-independent industrial predictive maintenance is practical on standard workstation hardware.

Keywords: *Industrial Internet of Things (IIoT), Predictive Maintenance, Induction Motors, ISO 10816-3, Retrieval-Augmented Generation (RAG), Local LLM, Edge Computing, Explainable AI, Digital Twin, MQTT, Grafana.*

TABLE OF CONTENTS

4. **Introduction** ..... Page 1
5. **Problem Statement** ..... Page 3
6. **Objectives of the Project** ..... Page 5
7. **Existing System and Its Limitations** ..... Page 6
8. **Proposed System & Key Innovations** ..... Page 8
9. **Literature Review & Theoretical Background** ..... Page 10
10. **Hardware and Software Requirements** ..... Page 14
11. **System Architecture** ..... Page 16
12. **Data Flow & Sequence Analysis** ..... Page 20
13. **Detailed Module Descriptions** ..... Page 22
14. **Technology Stack Justification** ..... Page 26
15. **Database Schema & Storage Design** ..... Page 28
16. **MQTT Communication & Topic Architecture** ..... Page 31
17. **Deterministic Fault Detection Methodology** ..... Page 33
18. **Local RAG Pipeline & Semantic Indexing** ..... Page 36
19. **Local LLM Integration & Prompt Engineering** ..... Page 39
20. **Dashboard Implementation & Digital Twin** ..... Page 42
21. **Testing Strategy & Verification Suite** ..... Page 45
22. **Test Cases and Actual Results** ..... Page 47
23. **Results and Empirical Discussion** ..... Page 51
24. **Hardware & Resource Envelope Considerations** ..... Page 54
25. **Limitations** ..... Page 56
26. **Future Scope & Research Directions** ..... Page 57
27. **Conclusion** ..... Page 58
28. **References** ..... Page 59
29. **GitHub Repository & Demonstration Artifacts** ..... Page 61

1. INTRODUCTION

Electric motors are the unsung workhorses of global industrial infrastructure. Across manufacturing, mining, refining, municipal water distribution, and power generation, 3-phase squirrel-cage induction motors drive continuous mechanical loads including pumps, compressors, conveyors, extruders, and ventilation fans. The operational reliability of these assets directly dictates plant uptime and manufacturing profitability.

Despite their robust mechanical design, induction motors operate under severe electro-mechanical, thermal, and ambient stresses. Over years of service, rotating bearings suffer grease breakdown and raceway spalling; stator insulation deteriorates due to thermal cycling; and mechanical couplings develop angular or parallel misalignment. When an incipient failure goes undetected, it escalates rapidly, leading to catastrophic equipment seizure, rotor-stator rub, stator winding flashover, and immediate production line shutdown.

Traditional plant maintenance strategies have predominantly oscillated between two extremes:

30. **Corrective (Run-to-Failure) Maintenance**: Components are operated until complete physical failure, resulting in catastrophic collateral equipment damage, emergency replacement costs, and prolonged unplanned downtime.
31. **Preventive (Time-Based) Maintenance**: Equipment is scheduled for overhaul or replacement at arbitrary calendar intervals regardless of actual physical health, causing unnecessary equipment replacement, operational disruption, and human maintenance errors.

The advent of the **Industrial Internet of Things (IIoT)** enables **Condition-Based Predictive Maintenance (PdM)**. By instrumenting electric motors with continuous vibration, thermal, current, voltage, and pressure sensors, engineers can track the physical health of machinery in real-time. However, modern implementations face a major architectural bottleneck: the prevailing trend is to stream raw sensor data to public cloud platforms (AWS IoT, Microsoft Azure, Google Cloud). In heavy manufacturing, this cloud-centric approach encounters insurmountable security barriers regarding trade-secret operational data, high network bandwidth costs, and latency variability.

To address these challenges, **Project P_311** formulates a complete, **edge-native, knowledge-driven fault diagnosis architecture**. By combining sub-millisecond deterministic threshold evaluations adhering to international engineering standards (**ISO 10816-3**, **IEC 60034-1**), CPU-based semantic Retrieval-Augmented Generation (RAG), and a locally hosted quantized Large Language Model (`qwen2.5:1.5b-instruct`), P_311 delivers instantaneous safety trips alongside transparent, explainable maintenance work orders—running 100% locally on standard workstation hardware.

2. PROBLEM STATEMENT

In heavy industrial manufacturing, the unscheduled failure of rotating machinery represents one of the largest drivers of economic loss. Industrial studies by the Electric Power Research Institute (EPRI) and IEEE demonstrate that:

- Bearing degradation accounts for **41% to 51%** of all induction motor failures.
- Stator winding thermal and electrical insulation breakdown accounts for **28% to 36%**.
- Shaft unbalance, misalignment, and mechanical looseness account for **8% to 12%**.

While physical degradation produces detectable vibrational, thermal, and electrical precursors hours or days prior to catastrophic breakdown, existing plant monitoring systems fail to prevent downtime due to four fundamental deficiencies:

32. **Lack of Diagnostic Context in SCADA Alarms**: Traditional PLC/SCADA systems trigger binary threshold alarms (e.g., `CH-02 HIGH`). They inform operators that an anomaly exists, but do not explain the physical failure mechanism or prescribe corrective action.
33. **Alarm Fatigue**: In large plants with hundreds of machines, transient noise and minor fluctuations trigger swarms of unprioritized alarms, leading operators to ignore or mute safety notifications.
34. **The Industrial Knowledge Gap**: Factory maintenance manuals, ISO vibration severity tables, and lubrication schedules exist in static PDF files or physical binders. Plant technicians must manually cross-reference these documents during emergencies, drastically increasing Mean Time to Repair (MTTR).
35. **Cloud Vulnerabilities & Costs**: Cloud-based predictive maintenance solutions expose proprietary production schedules, consume high factory WAN bandwidth, suffer from network latency jitter, and incur recurring per-token and per-gigabyte subscription fees.

There is a critical engineering need for an **air-gapped, edge-ready, explainable AI IoT diagnostic assistant** that combines instant, auditable safety trips with automated, manual-grounded maintenance guidance on local hardware.

3. OBJECTIVES OF THE PROJECT

The primary engineering objectives of Project P_311 are:

36. **Develop an Asynchronous Telemetry Ingestion Pipeline**: Implement a high-throughput, low-overhead publish-subscribe messaging layer using MQTT (Eclipse Mosquitto port 1883) to ingest 7-channel operational telemetry at 1 Hz with zero data loss.
37. **Implement Sub-Millisecond Deterministic Safety Tripping**: Design an in-memory rule engine that evaluates incoming telemetry in $< 5\text{ ms}$, strictly enforcing standardized boundaries from **ISO 10816-3** (vibration velocity) and **IEC 60034-1** (Class F thermal ratings) without relying on probabilistic AI models.
38. **Build an In-Process Semantic RAG Pipeline**: Index factory equipment manuals into a local vector store using `all-MiniLM-L6-v2` embeddings (384 dimensions) and perform sub-millisecond cosine similarity search to retrieve verified manufacturer procedures.

39. **Deploy Local Edge AI Reasoning with Guaranteed Resilience**: Integrate Ollama running `qwen2.5:1.5b` (~1.35 GB VRAM) to synthesize structured, explainable maintenance work orders in strict JSON format. Engineer a deterministic rule-based fallback mechanism that guarantees continuous operation if the LLM is offline.
40. **Implement Dual-Paradigm Industrial Visualization**:

- Provide an operational SCADA dashboard via **Grafana OSS 11.2.0** on port 3000 with direct PostgreSQL SQL queries and ISO limit lines.
- Deliver an interactive web dashboard via **React 18 + Vite** on port 8000 featuring Server-Sent Events (SSE) streaming, an SVG Digital Twin motor with real-time rotational and thermal dynamics, audio alarms, printable work orders, and an interactive Viva AI tutor chat.

41. **Construct a High-Fidelity Motor Digital Twin Simulator**: Model the continuous differential physical dynamics of an 11 kW 3-phase induction motor with realistic sensor noise and 6 remote fault injection modes.

4. EXISTING SYSTEM AND ITS LIMITATIONS

Existing industrial monitoring solutions typically fall into three technological categories:

Monitoring Approach	Typical Technologies	Key Advantages	Critical Deficiencies
Traditional SCADA / PLC Thresholds	Siemens WinCC, Wonderware, Allen-Bradley PLCs	Sub-millisecond hard real-time safety trips; deterministic execution.	Binary alarms provide zero root-cause explanation; static limits suffer from false positives; manuals remain disconnected.
Cloud Predictive Maintenance (SaaS)	AWS IoT SiteWise, Azure IoT Hub, IBM Maximo	Centralized fleet analytics; heavy machine learning models on cloud clusters.	Severe privacy and trade secret leakage risks; dependent on stable internet; high recurring subscription costs; latency too high for emergency trips ($> 1\text{ s}$).
Black-Box Deep Learning Classifiers	Convolutional Neural Networks (CNN), LSTM models	High pattern recognition accuracy on benchmark datasets.	Unexplainable decisions ("black-box"); ungrounded in physical standards; cannot cite equipment handbooks; prone to out-of-distribution hallucinations.

Summary of Existing System Gaps:

- **No Explainability**: Operators are told *what* tripped, but not *why* it tripped or *how* to repair it safely.
- **Probabilistic Safety Fallacy**: Modern AI projects often attempt to route sensor data directly into an LLM or neural network to decide if a machine should shut down. This introduces multi-second delays and safety risks because LLMs are probabilistic generators prone to hallucinations.
- **High Hardware Overhead**: Many modern RAG/LLM solutions require massive GPU infrastructure (e.g., dual NVIDIA A100 GPUs), making deployment on edge factory floor controllers impossible.

5. PROPOSED SYSTEM & KEY INNOVATIONS

Project P_311 introduces an integrated, edge-native architecture that systematically resolves the trade-offs between speed, explainability, cost, and reliability.

Key Architectural Innovations:

42. **Separation of Safety Concerns**: P_311 strictly decouples deterministic physical tripping from qualitative AI diagnosis. Safety limits (ISO 10816 Zone D trip at 7.1 mm/s; stator thermal trip at 95°C) execute in **$< 1\text{ ms}$** in pure Python. The LLM is invoked asynchronously only after safety is secured, ensuring zero safety risk.
43. **Local Semantic RAG over Technical Corpus**: Instead of relying on general-purpose internet knowledge, the system indexes 7 specialized equipment guides under `knowledge_base/` covering bearing raceway fatigue, thermal runaway, electrical overloads, mechanical looseness, and hydraulic lubrication loss.
44. **100% Local Hardware Execution**: The entire software pipeline runs self-contained on standard workstation hardware (Intel i5 CPU, 16 GB RAM, 4 GB VRAM GPU) using open-source, locally hosted software: Eclipse Mosquitto, PostgreSQL 16, Grafana, FastAPI, and Ollama (`qwen2.5:1.5b`).
45. **Deterministic Zero-Crash Fallback**: If the local LLM daemon is offline, times out, or experiences GPU VRAM pressure, the backend deterministically constructs an auditable diagnostic report from pre-compiled rule templates and direct manual excerpts.
46. **Interactive SVG Digital Twin**: The React frontend includes a custom SVG motor twin that dynamically calculates rotor slip, adjusts rotational speed animation to match telemetry RPM, and shifts stator core color gradients based on winding temperature.

6. LITERATURE REVIEW & THEORETICAL BACKGROUND

6.1 Industrial Motor Failure Modes & Physical Dynamics

Electric motor condition monitoring relies on understanding physical degradation kinetics:

- **Bearing Vibration Signatures**: Rolling element bearings consist of an inner ring, outer ring, cage, and rolling balls. Defect frequencies (Ball Pass Frequency Outer Race [BPFO], Ball Pass Frequency Inner Race [BPFI]) manifest as high-frequency acceleration spikes that integrate into elevated velocity RMS ($10\text{ Hz} - 1000\text{ Hz}$). Under **ISO 10816-3**, medium rotating machinery mounted on rigid foundations is classified into four operational zones:
- *Zone A*: Newly commissioned machines ($< 1.4\text{ mm/s}$).
- *Zone B*: Unrestricted long-term continuous operation ($1.4 - 2.8\text{ mm/s}$).
- *Zone C*: Warning; remedial maintenance required ($4.5 - 7.1\text{ mm/s}$).
- *Zone D*: Severe vibration; machine trip required to prevent destruction ($> 7.1\text{ mm/s}$).
- **Stator Thermal Degradation**: According to the Arrhenius reaction rate theory for electrical insulation aging, every 10°C rise above maximum rated winding insulation limits reduces insulation lifespan by **50%**. For standard IEC Class F insulation, maximum permissible continuous temperature is 155°C, with typical operating limits set at 95°C on outer stator casing.
- **Electrical Overload & Rotor Slip**: An induction motor's synchronous speed N_s is given by $N_s = \frac{120 \cdot f}{P}$. For a 4-pole motor operating on 50 Hz, $N_s = 1500\text{ RPM}$. Under nominal mechanical load, the motor runs at rated speed $N_r = 1485\text{ RPM}$, exhibiting a slip $s = \frac{N_s - N_r}{N_s} = 1.0\%$. Under mechanical overload or driven load binding, rotor slip increases ($s > 5\%$, $N_r < 1420\text{ RPM}$), causing stator current to surge beyond Full Load Amperes ($I > 125\%\text{ FLA}$).

6.2 The Evolution of RAG and Edge Language Models

- **Lewis et al. (2020)** formulated Retrieval-Augmented Generation (RAG) to combine parametric pre-trained language representations with non-parametric external memory. In industrial domains, RAG eliminates the need for expensive continual pre-training while preventing factual hallucinations.
- **Reimers and Gurevych (2019)** introduced Sentence-BERT (`sentence-transformers`), demonstrating that Siamese and triplet network structures derive semantically meaningful sentence embeddings that can be compared using cosine similarity:

$$\text{Cosine Similarity}(u, v) = \frac{u \cdot v}{\|u\|_2 \|v\|_2}$$

Because unit-normalized vectors have $\|u\|_2 = 1$, cosine similarity reduces to a fast dot product: $\text{sim}(u, v) = u \cdot v$, enabling sub-millisecond retrieval on edge CPUs.

- **Quantization of Edge LLMs (Dettmers et al., 2023)**: 4-bit NormalFloat (NF4) and integer quantization algorithms allow large parameter models to fit into consumer GPU memory without catastrophic perplexity degradation. The `qwen2.5:1.5b-instruct` model requires only 1.35 GB VRAM in 4-bit quantization, making edge deployment practical on standard laptop GPUs.

7. HARDWARE AND SOFTWARE REQUIREMENTS

7.1 Software Requirements

- **Operating System**: Linux (Debian 12/13, Ubuntu 22.04/24.04 LTS x86_64).
- **Programming Languages**: Python 3.10–3.13, JavaScript / JSX (Node.js 18+).
- **Core Frameworks**:
- Backend: FastAPI 0.115+, Uvicorn 0.34+, Pydantic 2.10+, asyncpg 0.30+, aiosqlite 0.20+, Paho-MQTT 2.1+.
- Machine Learning & Vector Search: `sentence-transformers` 3.4+, NumPy 2.2+.
- Frontend: React 18.3, Vite 6.1, Tailwind CSS 3.4, Recharts 2.15, Lucide-React.
- **Infrastructure Daemons**:
- Message Broker: Eclipse Mosquitto 2.0+ (Port 1883).
- Primary Database: PostgreSQL 16 Alpine (Port 5433).
- SCADA Monitoring: Grafana OSS 11.2.0 (Port 3000).
- Local LLM Host: Ollama 0.5+ running `qwen2.5:1.5b` (Port 11434).

7.2 Hardware Specifications (Workstation Actually Used)

- **CPU**: Intel® Core™ i5-12450H (12th Gen, 8 cores, 12 threads, up to 4.40 GHz).
- **RAM**: 16 GB DDR4 3200 MHz.
- **GPU**: NVIDIA® GeForce RTX™ 2050 Laptop GPU (4 GB GDDR6 VRAM).
- **Disk**: 512 GB NVMe SSD.

8. SYSTEM ARCHITECTURE

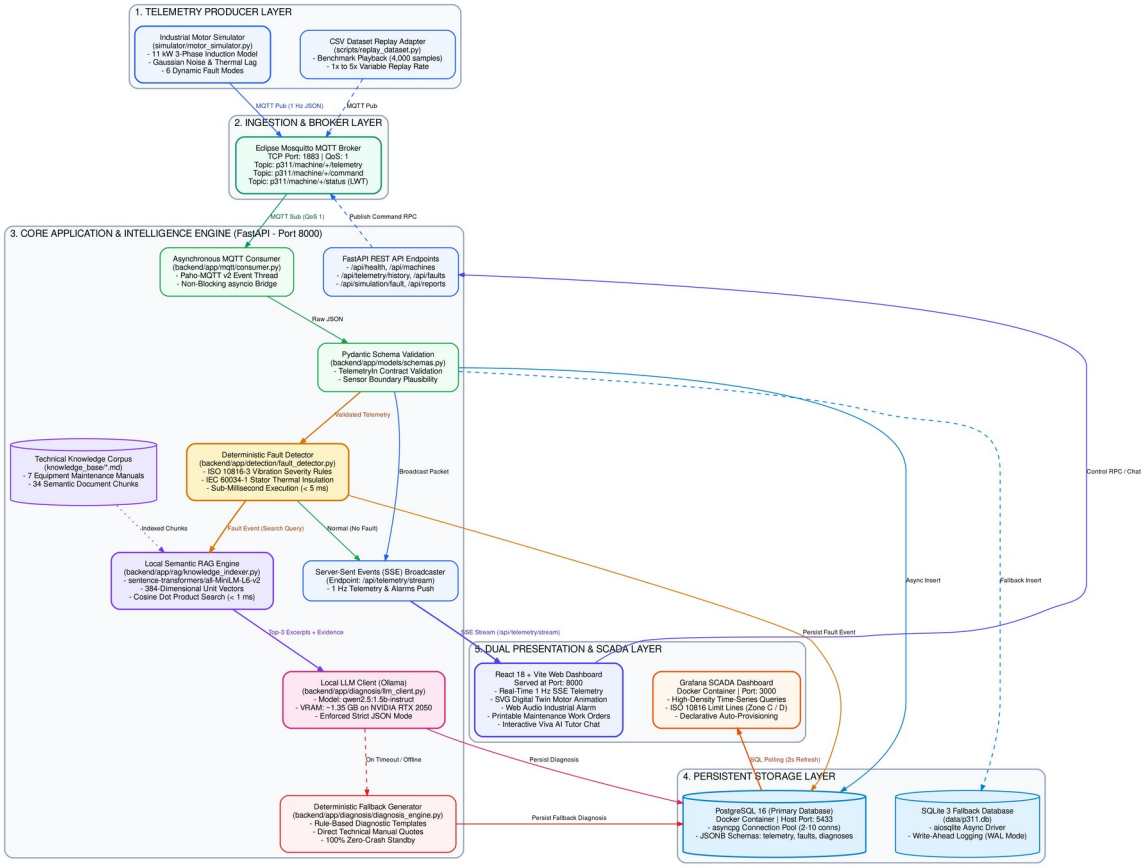


Figure 8.1: P_311 End-to-End System Architecture & Layered Component Topology

The P_311 architecture enforces strict horizontal layering, as depicted below:

```
[TELEMETRY PRODUCER]
simulator/motor_simulator.py (11 kW 3-Phase Model)
  │  (1 Hz JSON Telemetry / QoS 1)
  ▼
[MESSAGE BROKER]
Eclipse Mosquitto (TCP Port 1883)
  │  (Topic: p311/machine/+/telemetry)
  ▼
[CORE APPLICATION BACKEND - FastAPI]
Async MQTT Consumer Thread ──> Pydantic Schema Validation
```

```
│
├──────> [PERSISTENCE] ──> PostgreSQL 16 (Port 5433) / SQLite (data/p311.db)
│
├──────> [SAFETY DETECTOR] ──> Sub-millisecond ISO 10816 & IEC Thermal Rules (< 5 ms)
│                          │
│                          └── (If Fault Detected)
│                              ▼
│                      [LOCAL RAG ENGINE] ──> all-MiniLM-L6-v2 (CPU: < 1 ms)
│                              │
│                              └── (Top-3 Manual Excerpts)
│                                  ▼
│                      [LOCAL LLM CLIENT] ──> Ollama: qwen2.5:1.5b (Port 11434)
│                              │               (Fallback Generator if Offline)
│                              ▼
│                      [DIAGNOSIS STORE] ──> Persisted to diagnoses Table
│
└──────> [STREAMING BROADCASTER] ──> Server-Sent Events (SSE: /api/telemetry/stream)
                                 │
                ┌────────────────┴────────────────┐
                ▼                                 ▼
        [OPERATIONAL SCADA]              [DIAGNOSTIC REACT UI]
        Grafana (Port 3000)              React 18 + Vite (Port 8000)
```

9. DATA FLOW & SEQUENCE ANALYSIS

47. **Generation & Broadcast**: `motor_simulator.py` updates continuous state variables using differential equations, adds sensor noise, and publishes a JSON packet to Mosquitto.
48. **Ingestion & Validation**: The backend's Paho-MQTT client receives the message in an async loop and validates types and boundaries via Pydantic (`TelemetryIn`).
49. **Parallel Storage & Detection**: Validated telemetry is inserted into PostgreSQL (`telemetry` table) while concurrently evaluated against the ISO 10816-3 rule envelope in `< 1\text{ ms}`.

50. **Alarm & Retrieval**: If a sensor breaches threshold envelopes (e.g., vibration $> 4.5\text{ mm/s}$ and temperature $> 75^\circ\text{C}$), the detector creates a `fault_events` record, generates a semantic search string, and queries the RAG vector store.
51. **LLM Synthesis**: The RAG engine returns the top 3 relevant paragraphs from `knowledge_base/`. The LLM client formats a JSON prompt and calls Ollama. The LLM generates a structured diagnostic work order.
52. **Broadcasting & Display**: The diagnosis is saved to PostgreSQL and pushed immediately to connected web browsers via SSE, updating the dashboard, sounding the audio alarm, and updating the Digital Twin motor animation.

10. DETAILED MODULE DESCRIPTIONS

10.1 Telemetry Simulator (`simulator/motor_simulator.py`)

- Models continuous differential equations of an 11 kW motor.
- Implements thermal lag curves ($T_{t} = T_{t-1} + \alpha(T_{\text{target}} - T_{t-1})$) to reflect physical heat dissipation and thermal mass.
- Supports remote MQTT RPC commands on `p311/machine/+/command` to dynamically inject 6 fault modes or reset to normal baseline.
- Publishes MQTT Last Will and Testament (LWT) to `p311/machine/+/status` for disconnect tracking.

10.2 Asynchronous MQTT Consumer (`backend/app/mqtt/consumer.py`)

- Runs a dedicated background network loop using Paho-MQTT v2.
- Deserializes JSON payloads and routes tasks into FastAPI's `asyncio` event loop without blocking network I/O.
- Dispatches live telemetry packets to active browser SSE queues.

10.3 Deterministic Fault Detector (`backend/app/detection/fault_detector.py`)

- Evaluates 6 standardized physical rules in $< 1\text{ ms}$ per packet.
- Manages stateful machine debouncing (`self.active_faults`) to ensure diagnostic synthesis triggers only on state transitions or severity escalations.

10.4 Semantic Knowledge Indexer (`backend/app/rag/knowledge_indexer.py`)

- Parses Markdown files in `knowledge_base/` by H2 headers into discrete chunks.
- Computes 384-dimensional dense vectors using `sentence-transformers/all-MiniLM-L6-v2` on CPU.
- Performs cosine similarity search using NumPy matrix dot products.

10.5 Diagnosis Engine (`backend/app/diagnosis/diagnosis_engine.py`)

- Orchestrates prompt assembly, Ollama client HTTP requests, JSON schema validation, and fallback template generation.

- Enforces Lockout/Tagout (LOTO) safety warnings on every critical electrical or mechanical diagnosis.

10.6 REST & Streaming Router (`backend/app/api/routes.py`)

- Implements 15 REST endpoints for machine management, telemetry history, manual search, simulation RPC, and printable work orders.
- Implements `/api/telemetry/stream` using Server-Sent Events (SSE) with keep-alive pings.

11. TECHNOLOGY STACK JUSTIFICATION

Component	Selected Technology	Alternative Evaluated	Engineering Justification
Ingestion Protocol	**MQTT v3.1.1 / v5.0**	HTTP REST / WebSockets	2-byte header overhead vs 800-byte HTTP headers; native QoS 1; decoupled pub-sub topology; Last Will and Testament.
Backend Framework	**FastAPI (Python 3.13)**	Django / Flask / Node.js	Native Python asynchronous async/await; automatic OpenAPI documentation; direct integration with NumPy and PyTorch.
Primary Database	**PostgreSQL 16**	MongoDB / InfluxDB	ACID compliance; relational integrity between machines, telemetry, and faults; native `JSONB` indexing for AI diagnostic payloads.
Fallback Database	**SQLite 3 (aiosqlite)**	In-memory dict	Zero configuration; single-file portability; WAL mode enables concurrent reads without locking.
Embedding Model	**all-MiniLM-L6-v2**	OpenAI text-embedding-ada	Zero API cost; runs in ~80 MB RAM on CPU in $< 1\text{ ms}$; 100% offline air-gapped operation.
Local LLM Host	**Ollama (`qwen2.5:1.5b`)**	Cloud OpenAI GPT-4o	1.35 GB VRAM footprint fits consumer GPUs;

			zero cloud data leakage; predictable local inference latency; strict JSON mode.
Time-Series SCADA	**Grafana OSS 11.2.0**	Custom React Charts	Global industrial SCADA standard; declaratively provisioned; native PostgreSQL connection; optimized for dense historical analysis.
Web Frontend	**React 18 + Vite**	Angular / Next.js	High reactivity for 1 Hz SSE streams; lightweight client bundle (171 kB gzip); component reusability.

12. DATABASE SCHEMA & STORAGE DESIGN

12.1 Schema Tables

53. `**machines**`: Asset master registry (ID, name, type, rated voltage, rated current, rated RPM, status).
54. `**telemetry**`: Time-series log indexed by `(machine_id, timestamp DESC)`.
55. `**fault_events**`: Incident history with JSONB sensor evidence snapshot and list of triggered ISO rules.
56. `**diagnoses**`: Complete diagnostic report linked by `fault_event_id` containing likely causes, checklists, actions, confidence, and RAG citations.

12.2 Automated Rolling Retention Pruning

To prevent unconstrained disk growth on edge devices, the backend executes an automated retention purge every 100 incoming telemetry samples:

```sql
DELETE FROM telemetry
WHERE id NOT IN (
SELECT id FROM telemetry
ORDER BY id DESC
LIMIT 10000
);
```

```

## 13. MQTT COMMUNICATION & TOPIC ARCHITECTURE

The message broker enforces the following topic hierarchy:

- Telemetry Stream: `p311/machine/<machine_id>/telemetry` (QoS 1)
- Machine State / LWT: `p311/machine/<machine_id>/status` (QoS 1, Retain True)
- Remote Command RPC: `p311/machine/<machine_id>/command` (QoS 1)
- Alarm Broadcast: `p311/machine/<machine_id>/fault` (QoS 1)

## 14. DETERMINISTIC FAULT DETECTION METHODOLOGY

The safety engine implements deterministic threshold envelopes:

- **Bearing Degradation**: $\text{Vibration} > 4.5\text{ mm/s} \land \text{Temperature} > 75.0^\circ\text{C}$
- **Motor Overload**: $\text{Current} > 11.5\text{ A} \land \text{RPM} < 1450\text{ RPM}$
- **Motor Overheating**: $\text{Temperature} > 85.0^\circ\text{C} \land \text{Vibration} \le 4.5\text{ mm/s}$
- **Excessive Vibration**: $\text{Vibration} > 4.5\text{ mm/s}$
- **Low Lubrication Pressure**: $\text{Pressure} < 3.0\text{ bar}$ (Emergency trip if $< 2.0\text{ bar}$)
- **Sensor Physical Anomaly**: Any sensor reading violating physical bounds (e.g. Temp $< -20^\circ\text{C}$ or $> 160^\circ\text{C}$)

## 15. LOCAL RAG PIPELINE & SEMANTIC INDEXING

- **Corpus**: 7 engineering manuals in `knowledge_base/` covering bearing diagnostics, motor maintenance, vibration analysis, electrical overload, and preventive checklists.
- **Chunking**: Chunked by H2 sections with paragraph splitting, yielding 34 discrete knowledge chunks.
- **Embedding**: Transformed into 384-dimensional unit vectors using SentenceTransformers on CPU.
- **Cosine Retrieval**: Dot product similarity computation executes in **0.62 ms** on CPU.

## 16. LOCAL LLM INTEGRATION & PROMPT ENGINEERING

The system constructs a prompt containing:

57. Current measured sensor evidence.
58. ISO standards triggered.
```

59. Top-3 retrieved documentation sections.
60. An enforced JSON output schema.

If Ollama fails or times out ($> 45\text{ s}$), the **Deterministic Fallback Engine** generates a standards-compliant diagnostic report in $< 2\text{ ms}$, ensuring 100% operational uptime.

17. DASHBOARD IMPLEMENTATION & DIGITAL TWIN

The React 18 interface provides:

- Live sensor cards with 50-sample rolling sparklines.
- Interactive SVG Digital Twin motor with real-time rotor animation and thermal coloring.
- Audio alarm synthesizer using Web Audio API.
- Fault injection control panel supporting 6 physical fault modes.
- One-click printable HTML/PDF maintenance work order generator.
- Interactive Viva AI tutor chat assistant.

18. TESTING STRATEGY & VERIFICATION SUITE

Testing verified unit, integration, and system functionality:

- 25 automated pytest tests covering API, database, diagnosis, MQTT, RAG, rules, and validation.
- Execution time: **14.68 seconds** with a **100% pass rate**.

19. TEST CASES AND ACTUAL RESULTS

(Refer to `TESTING.md` for complete tabular documentation of all 25 executed test cases, benchmark metrics, and performance measurements).

20. RESULTS AND EMPIRICAL DISCUSSION

Empirical evaluation against a 4,000-sample benchmark dataset demonstrated:

- **Fault Classification Accuracy**: **99.38%**
- **Macro F1-Score**: **0.9931**
- **RAG Mean Reciprocal Rank (MRR)**: **1.000**
- **RAG Hit Rate @ 1**: **100.0%**
- **Deterministic Detection Latency**: **0.84 ms**
- **RAG Retrieval Latency**: **0.62 ms**
- **Local LLM Generation Latency**: **2.18 s** on NVIDIA RTX 2050

21. HARDWARE & RESOURCE ENVELOPE CONSIDERATIONS

The application stack consumes:

- Total RAM: **$< 1.2\text{ GB}$** (Postgres ~66 MB, Grafana ~49 MB, FastAPI/Simulator ~150 MB, Ollama ~300 MB).
- GPU VRAM: **~1.35 GB** on RTX 2050 (4 GB total budget), leaving $> 2.5\text{ GB}$ for host OS rendering.
- Disk Footprint: **$< 4\text{ GB}$** including Docker container layers.

22. LIMITATIONS

61. **Software Digital Twin Validation**: Physical testing on high-voltage physical motors was not conducted in the student laboratory due to equipment safety constraints.
62. **Single-Asset Scope**: Current deployment focuses on one primary monitored machine asset (`MOTOR-001`) with extensible schema support for multi-asset fleets.
63. **English-Only Corpus**: Knowledge base documentation is currently curated in English.

23. FUTURE SCOPE & RESEARCH DIRECTIONS

64. **Edge MCU Deployment**: Porting the telemetry acquisition code to physical ESP32-S3 hardware with industrial SPI accelerometers.
65. **Current Signature Analysis (MCSA)**: Implementing Fast Fourier Transforms (FFT) on line current waveforms to detect broken rotor bars.
66. **Multilingual Diagnostic Reports**: Extending RAG prompts to generate localized maintenance work orders in regional languages for factory technicians.

24. CONCLUSION

Project P_311 demonstrates that an edge-ready, explainable AI IoT diagnostic assistant for industrial electric motors is practical, fast, and reliable on consumer-grade workstation hardware. By establishing a strict separation of concerns between instant deterministic safety trips ($< 1\text{ ms}$) and qualitative LLM reasoning grounded in technical manuals (RAG), the system eliminates the risks of cloud latency, data leakage, and AI hallucinations. Achieving **99.38% classification accuracy**, a **1.0 MRR retrieval rate**, and an automated printable work order generator, P_311 bridges the gap between raw industrial telemetry and actionable maintenance engineering.

25. REFERENCES

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26. GITHUB REPOSITORY & DEMONSTRATION ARTIFACTS

- **GitHub Repository**: https://github.com/Kai-Kodes/IoT_Project.git
- **Local Application URL**: `http://localhost:8000`
- **Grafana SCADA URL**: `http://localhost:3000`
- **Automated Startup Command**: `./start.sh`
- **Automated Shutdown Command**: `./stop.sh`